AIML PYTHON MINI PROJECT SUBMISSION

Name of student: Vaibhav Vijay Kate

Domain/subject name: AIML (Artificial Intelligence & Machine Learning)

College and Branch name: Government Polytechnic College Washim (IF)

Project 5: Inventory Management System

Code: -

```
# Inventory Management System


inventory = {}


# Add Product

def add_product():


    product_id = input("Enter Product ID: ")


    if product_id in inventory:

        print("Product ID Already Exists!")

        return


    name = input("Enter Product Name: ")

    category = input("Enter Category: ")

    price = float(input("Enter Price: "))

    quantity = int(input("Enter Quantity: "))

    reorder_level = int(input("Enter Reorder Level: "))
```

```
    inventory[product_id] = {
        "name": name,
        "category": category,
        "price": price,
        "quantity": quantity,
        "reorder_level": reorder_level
    }

    print("Product Added Successfully!")

# Stock In
def stock_in():

    product_id = input("Enter Product ID: ")

    if product_id in inventory:

        qty = int(input("Enter Quantity to Add: "))

        inventory[product_id]["quantity"] += qty
```

```
        print("Stock Updated Successfully!")

    else:
        print("Product Not Found")

# Stock Out
def stock_out():

    product_id = input("Enter Product ID: ")

    if product_id in inventory:

        qty = int(input("Enter Quantity to Remove: "))

        if qty <= inventory[product_id]["quantity"]:

            inventory[product_id]["quantity"] -= qty

            print("Stock Removed Successfully!")
```

```
        else:

            print("Not Enough Stock!")


    else:

        print("Product Not Found")


# View Inventory

def view_inventory():


    if len(inventory) == 0:

        print("No Products Available")

        return


    print("\n===== INVENTORY =====")


    for pid, product in inventory.items():


        print("\nProduct ID :", pid)

        print("Name :", product["name"])

        print("Category :", product["category"])

        print("Price :", product["price"])
```

```
        print("Quantity :", product["quantity"])

        print("Reorder Level :", product["reorder_level"])


# Low Stock Alert

def low_stock_alert():


    print("\n===== LOW STOCK PRODUCTS =====")


    found = False


    for pid, product in inventory.items():


        if product["quantity"] <= product["reorder_level"]:


            found = True


            print(pid, "-", product["name"])


    if found == False:

        print("No Low Stock Products")
```

```
# Generate Report
def generate_report():

    total_products = len(inventory)

    total_value = 0

    for product in inventory.values():

        total_value += (
            product["price"] * product["quantity"]
        )

    print("\n===== INVENTORY REPORT =====")
    print("Total Products :", total_products)
    print("Total Stock Value :", total_value)

# Main Menu
while True:

    print("\n===== INVENTORY MANAGEMENT SYSTEM =====")
```

```
print("1. Add Product")

print("2. Stock In")

print("3. Stock Out")

print("4. View Inventory")

print("5. Low Stock Alert")

print("6. Report")

print("7. Exit")


choice = input("Enter Choice: ")


if choice == "1":

    add_product()


elif choice == "2":

    stock_in()


elif choice == "3":

    stock_out()


elif choice == "4":

    view_inventory()
```

```
elif choice == "5":
    low_stock_alert()

elif choice == "6":
    generate_report()

elif choice == "7":
    print("Thank You!")
    break

else:
    print("Invalid Choice")
```

Output Screenshort: -

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    TEST RESULTS    PORTS        Python: inventory_management

> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe c:/Users/sdf/Desktop/ITR2026/Projects/inventory_management.py

===== INVENTORY MANAGEMENT SYSTEM =====
1. Add Product
2. Stock In
3. Stock Out
4. View Inventory
5. Low Stock Alert
6. Report
7. Exit
Enter Choice: 1
Enter Product ID: 101
Enter Product Name: Samsung M34 5G
Enter Category: Moible
Enter Price: 19500
Enter Quantity: 12
Enter Reorder Level: 5
Product Added Successfully!

===== INVENTORY MANAGEMENT SYSTEM =====
1. Add Product
2. Stock In
3. Stock Out
4. View Inventory
5. Low Stock Alert
6. Report
7. Exit
Enter Choice: 2
Enter Product ID: 101
Enter Quantity to Add: 3
Stock Updated Successfully!
```

```
===== INVENTORY MANAGEMENT SYSTEM =====
1. Add Product
2. Stock In
3. Stock Out
4. View Inventory
5. Low Stock Alert
6. Report
7. Exit
Enter Choice: 3
Enter Product ID: 101
Enter Quantity to Remove: 2
Stock Removed Successfully!

===== INVENTORY MANAGEMENT SYSTEM =====
1. Add Product
2. Stock In
3. Stock Out
4. View Inventory
5. Low Stock Alert
6. Report
7. Exit
Enter Choice: 4

===== INVENTORY =====

Product ID : 101
Name : Samsung M34 5G
Category : Moible
Price : 19500.0
Quantity : 13
Reorder Level : 5
```

```
===== INVENTORY MANAGEMENT SYSTEM =====
1. Add Product
2. Stock In
3. Stock Out
4. View Inventory
5. Low Stock Alert
6. Report
7. Exit
Enter Choice: 5

===== LOW STOCK PRODUCTS =====
No Low Stock Products

===== INVENTORY MANAGEMENT SYSTEM =====
1. Add Product
2. Stock In
3. Stock Out
4. View Inventory
5. Low Stock Alert
6. Report
7. Exit
Enter Choice: 6

===== INVENTORY REPORT =====
Total Products : 1
Total Stock Value : 253500.0
```

```
===== INVENTORY MANAGEMENT SYSTEM =====
1. Add Product
2. Stock In
3. Stock Out
4. View Inventory
5. Low Stock Alert
6. Report
7. Exit
Enter Choice: 6

===== INVENTORY REPORT =====
Total Products : 1
Total Stock Value : 253500.0

===== INVENTORY MANAGEMENT SYSTEM =====
1. Add Product
2. Stock In
3. Stock Out
4. View Inventory
5. Low Stock Alert
6. Report
7. Exit
Enter Choice:
```